FCMMA: Fourier Conditional Mask-Based Mixed Attention Method for Hyperspectral Anomaly Detection

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***Abstract*— In recent years, reconstruction-based methods have achieved excellent detection results in the field of hyperspectral anomaly detection (HAD). These methods predominantly operate on two aspects regarding their working principles: 1) reconstructing background pixels and 2) suppressing anomalous pixels. However, most methods only tackle the HAD task from the spatial and spectral domains, making it challenging to effectively suppress anomalies. To eliminate these issues, this article proposes a Fourier conditional mask-based mixed attention (FCMMA) method. First, we propose the FCMMA method for HAD. FCMMA generates a conditional mask (CMASK) that suppresses anomalous high-frequency information and preserves background low-frequency information in the frequency domain, optimizing the anomaly detection process. In addition, to achieve fine-grained HAD, we propose the Fourier anomaly suppression filter (FASF). FASF uses Fourier techniques to manage background and anomalies, improving detection via precise frequency decoupling. Finally, a CMASK network is designed to effectively suppress anomalies. The CMASK network integrated the FASF module and the spatial–spectral multilayer perceptual (SSMLP) machine module together to enhance the transformation and representation capabilities of the generated masks, which can also help suppress anomalies. The results on five different datasets show that the proposed method is more effective and superior when compared to nine state-of-the-art methods.**

***Index Terms*— Anomaly detection, conditional mask (CMASK), Fourier transform, hyperspectral images (HSIs).**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain spatial texture information in 2-D and rich spectral details, enabling precise differentiation of distinct substances' characteristics [1], [2], [3]. The integration of spatial and spectral data in hyperspectral technology brings significant benefits. In agriculture, it aids in precision farming. In environmental monitoring, it helps detect changes over time. In mineral exploration, it identifies subtle geological signatures [4].

In the field of hyperspectral imagery, hyperspectral anomaly detection (HAD) is crucial [5]. It focuses on identifying pixels with spectral properties that starkly contrast with the background. Unlike target detection, HAD does not depend on prior spectral information [6]. Because HAD operates without prior spectral information, it finds wide application in military surveillance, intelligent agriculture, search and rescue operations, and other diverse fields [7], [8], [9]. Meanwhile, HAD is also regarded as an unsupervised binary classification task with detection difficulties such as an imbalance between the number of background and anomalous pixels and no a priori spectral information [10].

In the initial stage of HAD, statistics-based methods are usually used. These methods assume a specific data distribution for the background data, identifying pixels as abnormal if they deviate from this distribution [11]. The Reed–Xiaoli (RX) detector [12] assumes that the background follows a Gaussian multivariate distribution. It detects anomalies by calculating the Mahalanobis distance between the measured pixels and the background estimation model, which is derived from the covariance matrix calculation [13]. However, the fundamental RX algorithm is not appropriate for intricate background distributions, leading to the development of several enhanced techniques [14], [15], [16]. For instance, local RX (LRX) [17] enhances background estimation by employing a local window around the test pixel for more comprehensive coverage. Liu and Chang [18] employed a multiwindow strategy to accommodate abnormal targets of varying scales, and the kernel RX (KRX) [19] further enhances the separability of background and anomalies by transforming the original data into kernel space. Recently, Sun et al. [20] proposed an information entropy estimation based on point-set topology (IEEPST) algorithm, which integrates point-set topology with information entropy to enhance the accuracy and efficiency of land cover extraction and anomaly detection.

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However, assumptions of ideal and singular distributions often fail to capture the complex distribution of land backgrounds in reality [21]. To address this issue, representation-based approaches have been proposed. These methods are based on the weak assumption that the background can be represented by several base signals extracted from HSIs, whereas anomalies cannot [22], [23]. For example, the collaborative representation-based detector (CRD) [24] treats each background pixel as a linear combination of its neighboring pixels, categorizing pixels that cannot be effectively represented by this combination as anomalies. The background joint sparse representation detection (BJSRD) [25] method assumes that anomalies in each pixel can be detected through a sparse representation, by using only a few coefficients from a specific base or dictionary. Low-rank representation (LRR) [26] identifies anomalies by subtracting the reconstructed background from the original image after finding the representation with the lowest rank.

However, representation-based methods utilize only the underlying features and require manual tuning of the parameters, which creates uncertainty in the detection results [27], [28], [29]. With the development of deep learning technology, DNN-based methods have become a mainstream solution to the HAD problem [30]. Jiang et al. [31] proposed an unsupervised low-rank embedded network (LREN), in which the acquired low-rank representation is continuously optimized to fit a specific density estimation through the joint training of an autoencoder (AE) and a Gaussian mixture model. Distinguishing from the above approaches that use the AE only as a feature extractor, Fan et al. [32] proposed the robust graph AE (RGAE) detector. This method directly uses AE as a driver to better improve detection by embedding a superpixel segmentation-based graph regularization term (SuperGraph) into AE to obtain a resultant graph that is more robust to noise and anomalies.

Building upon these advancements, Wang et al. [33] proposed an autonomous HAD network (Auto-AD). This method improves error map contrast with an adaptive loss function that reduces anomaly weights during training. Xiang et al. [11] further developed AE and proposed a guided AE-based detector (GAED). The method uses a simple seven-layer AE network to bootstrap the image by calculating the spectral similarity, enabling the suppression of anomalous pixels. Wu and Wang [34] proposed a transformer (TR)-based AE framework (TAEF), which takes into account both local spatial information and transmitted global spatial information. By using the Transformer [35], [36] as an encoder, higher order spatial features can be accurately characterized and reconstructed more efficiently from the background of the HSI. Cheng et al. [37] recently proposed the deep feature aggregation network (DFAN), which can effectively utilize spectral and spatial information to enhance background and anomaly representations for improved HAD.

The rise of large models has infused new vitality into remote sensing image processing, and artificial intelligence models such as SpectralGPT [38], PASSNET [39], VBIGAN-AD [40], and the position-aware graph-convolutional neural network [41] have achieved breakthroughs in hyperspectral remote sensing. The development of these models has significantly advanced algorithmic techniques, fostering key progress in detecting, classifying, identifying, quantifying, and characterizing objects and features in captured data [42].

From the above studies, it is clear that most current HAD methods rely on encoding networks, particularly those such as convolutional AEs and Transformers, which work in concert to accurately reconstruct background information and effectively suppress anomalous information [43], [44], [45]. However, these reconstruction methods might have some limitations in feature extraction [46]. First, these methods only capture anomaly information in the spatial and spectral domains, and the suppression of anomalies is usually rough. They ignore the role of frequency-domain information in suppressing anomalous effects. [47]. Second, most methods reconstruct the background information using dimensionality reduction methods to reduce the computational pressure, and this may lead to the loss of some background information [30]. Finally, most existing methods use fully convolutional networks for image reconstruction, but these methods suffer from the problem of ignoring external information of the convolutional kernel due to their small perceptual domain [48], [49].

To address the above problems, a Fourier conditional mask-based mixed attention (FCMMA) method is proposed for HAD. During the training phase, anomalies are suppressed comprehensively and accurately by finely distinguishing anomalous pixels from background pixels in the frequency domain, while retaining more background information. First, the detailed structure integrates the conditional mask (CMASK) for anomaly suppression with the original image through channel connectivity to form a hybrid image condition. Subsequently, the mixed image condition was selected for filtering by using the mixed attention weight discriminant (MAWD) block. In addition, to achieve fine-grained suppression of anomalous pixels, a Fourier anomaly suppression filter (FASF) is proposed in this article. The block uses the Fourier technique to effectively deal with low-frequency background and high-frequency anomaly components, and coarse and fine suppression of anomalies through frequency decoupling. Finally, to be able to effectively adjust the reconstruction tendency during the image reconstruction process, this article proposes a CMASK network for generating a CMASK that adequately suppresses anomalies. The CMASK network first uses FASF to fully suppress anomalies in the frequency domain, and then further enhances the transform and representation capabilities of the generated masks through the spatial–spectral multilayer perceptual (SSMLP) machine module to improve its performance in both spectral and spatial dimensions.

In summary, the main contributions of this article can be summarized as follows.

1) We propose an FCMMA method for HAD. FCMMA can generate a CMASK that guides the network to suppress anomalous high-frequency information and preserving background low-frequency information in the frequency domain, thereby guiding the anomaly detection process to optimal performance.

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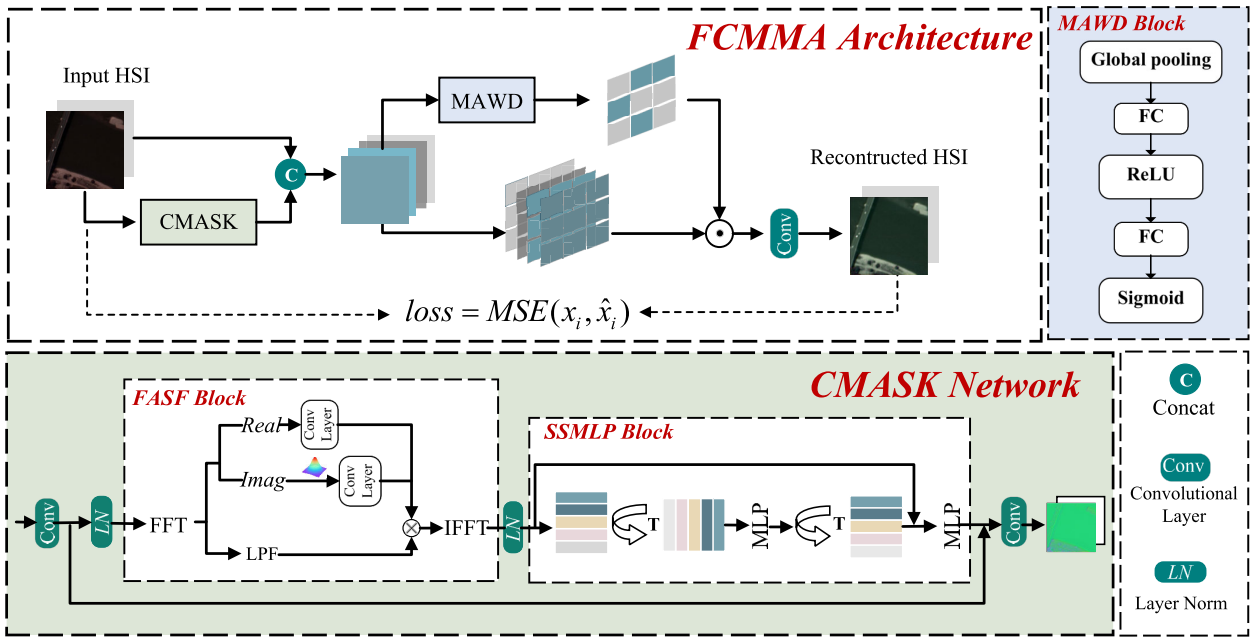


Fig. 1. Overall pipeline for the proposed FCMMA. First, the original HSI is spliced into the conditional image in the channel dimension with the image that is sufficiently anomaly suppressed in the frequency domain after going through the CMASK network. Second, the reconstructed HSI with sufficient anomaly suppression is obtained by multiplying the conditioned image with its filtered image after MAWD-weighted channel selection. Finally, the loss is calculated by comparing the reconstructed image with the original image.

2) An FASF is proposed, which can efficiently and accurately process the anomalous components in the frequency domain using Fourier techniques. FASF achieves fine suppression of anomalies through frequency decoupling, which makes the anomaly suppression more comprehensive and precise and significantly improves the detection performance.
3) A CMASK network is proposed for generating CMASKs that adequately suppress anomalies. The generated mask can effectively guide the reconstruction trend of different pixels in the image and enhance the reconstruction expression of background pixels while suppressing the reconstruction of anomalous pixels.

The rest of this article is organized as follows. Section II presents the detailed architecture of the proposed FCMMA. Section III presents the experimental results and analyses. Finally, Section IV concludes this article.

II. Methodology

The overall diagram of the proposed method FCMMA is shown in Fig. 1. FCMMA performs anomaly suppression and background expression enhancement via the CMASK network and obtains better background reconstruction results through weighted attention channel selection. Among them, the CMASK network contains two important blocks, FASF and SSMLP. In addition, weighted attention channel selection is implemented through the MAWD block.

A. Overall Framework

Given an HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, H, W, and C represent its height, width, and the number of bands, respectively. The HSI cube $\mathbf{X}$ is fed into the proposed network for reconstruction, and the goal is to reconstruct a pure background image.

The CMASK $\mathbf{M} \in \mathbb{R}^{H\times W\times C}$ is the fully suppressed anomaly mask image, which is obtained after the input $\mathbf{X}$ is passed through the CMASK network. The mask $\mathbf{M}$ is used to direct the image to pay more attention to the background information during the reconstruction process and suppress the expression of anomalies. $\mathbf{M}$ is obtained by

$$\mathbf{M} = \mathrm{CMASK}(\mathbf{X}) \tag{1}$$

where CMASK$(\cdot)$ is a function of the CMASK network. As shown in Fig. 1, the network consists of two main parts: the FASF block and the SSMLP block.

There are a total of four steps in the process of obtaining a mask by the CMASK network.

1) $\mathbf{X}$ needs to first go through a convolutional layer that adjusts the number of channels to C_1 and a normalization layer to obtain the internal network input $\mathbf{M}_I \in \mathbb{R}^{H\times W\times C_1}$

$$\mathbf{M}_I = \mathrm{LN}(\mathrm{conv}(\mathbf{X})) \tag{2}$$

where conv$(\cdot)$ is the convolutional layer and LN$(\cdot)$ is the layer normalization.
2) To suppress the expression of anomalies, $\mathbf{M}_I$ is fed into the FASF block to obtain the output $\mathbf{M}_{\mathrm{FASF}} \in \mathbb{R}^{HW\times C_1}$

$$\mathbf{M}_{\mathrm{FASF}} = \mathrm{FASF}(\mathbf{M}_I) \tag{3}$$

where FASF$(\cdot)$ is the FASF module.
3) $\mathbf{M}_{\mathrm{FASF}}$ is fed into the SSMLP block after inputting the normalization layer to improve the expressiveness of the generated mask and further improve the performance of anomaly detection

$$\mathbf{M}_{\mathrm{SSMLP}} = \mathrm{MLP}\left(\mathrm{MLP}\left(\mathrm{LN}(\mathbf{M}_{\mathrm{FASF}})^{\mathrm{T}}\right)^{\mathrm{T}}\right) + \mathrm{LN}(\mathbf{M}_{\mathrm{FASF}}) \tag{4}$$

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where $\mathbf{M}_{\text{SSMLP}} \in \mathbb{R}^{H\times W\times C_1}$, and MLP$(\cdot)$ is a function of the multilayer perceptual machine [50].

4) The final mask $\mathbf{M} \in \mathbb{R}^{H\times W\times C}$ is obtained by regressing the original number of channels through one more convolutional layer

$$\mathbf{M} = \text{conv}(\mathbf{M}_{\text{SSMLP}}). \tag{5}$$

Afterward, the original HSI $\mathbf{X}$ is spliced with the CMASK $\mathbf{M}$ through the channel dimension to form the conditional image $\mathbf{X}_{\text{cat}} \in \mathbb{R}^{H\times W\times 2C}$. $\mathbf{X}_{\text{cat}}$ is used as a conditioned image and contains two important pieces of information. These are the original HSI information and the mask information that is used to suppress anomalous conditions. $\mathbf{X}_{\text{cat}}$ is formulated as

$$\mathbf{X}_{\text{cat}} = \text{concatenate}(\mathbf{X}, \mathbf{M}) \tag{6}$$

where concatenate$(\cdot)$ is the splice operation.

After acquiring the conditional image $\mathbf{X}_{\text{cat}}$, it is passed through the MAWD network to obtain the attention weight filter $\mathbf{W} \in \mathbb{R}^{H\times W\times 2C}$. This process is an adaptive one, where $\mathbf{W}$ is employed to filter out anomalous information and is continuously updated through successive iterations. It simultaneously retains comprehensive background details. The MAWD network can adjust the weight filters based on the input image's features, ensuring effective information filtering through this adaptive mechanism. The formula for $\mathbf{W}$ is

$$\mathbf{W} = \text{MAWD}(\mathbf{X}_{\text{cat}}) \tag{7}$$

where MAWD$(\cdot)$ is a function of the MAWD network.

Multiplying the attention weighting filter $\mathbf{W}$ with $\mathbf{X}_{\text{cat}}$ gives the resultant image after weighted channel selection. After this, the reconstructed image $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times C}$ is obtained by adjusting the channel dimensions through a single-layer convolution operation. The $\hat{\mathbf{X}}$ produces both better context and highly suppressed anomalous information, and the formula is

$$\hat{\mathbf{X}} = \text{conv}(\mathbf{W} \odot \mathbf{X}_{\text{cat}}) \tag{8}$$

where conv$(\cdot)$ is a single layer convolution operation. $\odot$ denotes the Hadamard product, which is also known as elementwise multiplication.

Finally, the loss $\mathcal{L}$ is obtained by calculating the reconstruction error matrix

$$\mathcal{L} = \sum_{i=1}^{H}\sum_{j=1}^{W}\left\|\mathbf{X}_{i,j,:} - \hat{\mathbf{X}}_{i,j,:}\right\|_2^2. \tag{9}$$

Overall, anomaly detection in FCMMA is divided into two main phases. First, there is a training phase, where the original HSI $\mathbf{X}$ is used as the network input and the whole network is continuously optimized by the loss function $\mathcal{L}$. Next is the detection phase, which is after the completion of the training phase, and the difference map is obtained by calculating the difference between the reconstructed image and the original image as the detection result.

B. FASF Block

The FASF block is used to effectively suppress anomalies in the generated CMASK. In this block, the Fourier transform is used to shift image processing to the frequency domain. This shift allows us to fine-tune the suppression of anomalies using frequency decoupling. Through this process, the anomalies in the real and imaginary parts of the spectrum are individually and precisely suppressed, thus improving the overall effectiveness of anomaly suppression in conditional masking.

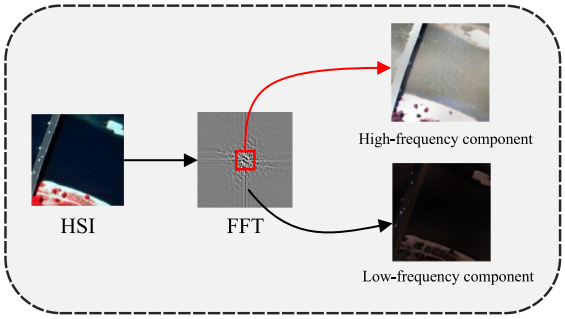


Fig. 2. Schematic of the high-frequency part of the image and the low-frequency part of the image obtained by Fourier transform of the HSI.

In this block, anomaly suppression is performed based on the fact that the high-frequency and low-frequency parts of the image have different characteristics, respectively. Anomalies usually exhibit a high-frequency component in the frequency domain [51]. This is because anomalies usually manifest themselves as sharp localized changes or mutations in the image that correspond to high-frequency information in the frequency domain [52].

As shown in Fig. 2, the high-frequency part and the low-frequency part of the HSI are obtained by performing Fourier transform and inverse transform. It can be seen that the low-frequency part contains the overall structure, contours, and large-scale variations of the image. In contrast, the high-frequency part contains image details, textures, and edge information [53], [54]. Therefore, anomaly suppression is to screen and process the high-frequency portion of the image in the frequency domain.

Given the inputs to the FASF block as $\mathbf{M}_I \in \mathbb{R}^{H\times W\times C_1}$, first, $\mathbf{M}_F = F(\mathbf{M}_I)$ is obtained by the fast Fourier transform (FFT). $F(\cdot)$ is the FFT function, which is expressed as

$$\mathbf{M}_F(u, v) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1}\mathbf{M}_I(h, w)\cdot e^{-2\pi i(uh+vw)} \tag{10}$$

where H and W represent the height and width of the HSI, respectively.

From the above equation, features at different spatial locations in the $\mathbf{M}_F$ correspond to different frequency components of the $\mathbf{M}_I$, and this process converts $\mathbf{M}_I$ to $\mathbf{M}_F$ [51].

Next, coarse-level and fine-level anomaly suppression operations are performed on the $\mathbf{M}_F$ by two different branches.

In branch one, a low-pass filter (LPF) [55] is used to filter out the high-frequency components of the $\mathbf{M}_F$. This step is a coarse filtering of the anomalies in $\mathbf{M}_F$. The coarse suppression result for this branch is given as

$$\mathbf{M}_{\text{coarse}} = \text{LPF}(\mathbf{M}_F). \tag{11}$$

The specific formula for LPF$(\cdot)$ is

$$H(u, v) = \begin{cases} 1, & \text{if } u^2 + v^2 \le D_0 \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

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where D_0 is the cutoff frequency of the LPF and (u, v) are the coordinates in the frequency domain.

In the other branch, the complex feature $\mathbf{M}_F$ is split into a real part $\mathbf{M}_{\text{real}}$ and an imaginary part $\mathbf{M}_{\text{imag}}$. The formula is

$$\mathbf{M}_F = \mathbf{M}_{\text{real}} + i \cdot \mathbf{M}_{\text{imag}}. \tag{13}$$

In the frequency domain, the real part is usually associated with the low-frequency component of the image, which can provide information about the structure of the image, such as the smoothness of the image [56]. However, the anomalous features tend to manifest themselves as high-frequency components that may not be readily apparent in the real part. On the contrary, the imaginary part contains the frequency-dependent phase information of the image and describes the vertical component of the image, which can reflect the anomalous detail features to some extent [57]. Therefore, extracting features in-depth while suppressing the expression of anomalous features requires different processing of the real and imaginary parts, respectively.

Convolutional operations are used on the real part $\mathbf{M}_{\text{real}}$ to fuse global features for deep background feature extraction. The formula is

$$\mathbf{M}'_{\text{real}} = \mathbf{K} * \mathbf{M}_{\text{real}} \tag{14}$$

where $\mathbf{K}$ is an ordinary convolution kernel and $*$ is the convolution operator.

Gaussian convolution is used for the imaginary part $\mathbf{M}_{\text{imag}}$ to smooth the vertical component of the image and effectively suppress anomalous information. The formula is

$$\mathbf{M}'_{\text{imag}} = \mathbf{G} * \mathbf{M}_{\text{imag}} \tag{15}$$

where $\mathbf{G}$ is a 2-D Gaussian convolution kernel.

Gaussian convolution can effectively smooth the image and reduce the noise in the image [58]. It also has the function of low-pass filtering, which provides good suppression of high-frequency anomalies while retaining low-frequency image structure for finer anomaly suppression [59]. The 2-D Gaussian function $\mathbf{G}(x, y)$ is given as

$$\mathbf{G}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right). \tag{16}$$

The fine anomaly suppression result $\mathbf{M}_{\text{fine}}$ is obtained in this branch. The formula is

$$\mathbf{M}_{\text{fine}} = \mathbf{M}'_{\text{real}} + i \cdot \mathbf{M}'_{\text{imag}}. \tag{17}$$

Finally, multiply the coarse anomaly suppression results from the two branches. Then, perform an inverse FFT (IFFT) to obtain the anomaly suppression result of the FASF block, which is $\mathbf{M}_I = F^{-1}(\mathbf{M}_F)$. $F^{-1}(\cdot)$ is a function of the IFFT, which is expressed as

$$\mathbf{M}_I(h, w) = \frac{1}{HW} \sum_{u=0}^{H-1} \sum_{v=0}^{W-1} \mathbf{M}_F(u, v) \cdot e^{2\pi i(uh+vw)} \tag{18}$$

where H and W represent the height and width of the HSI, respectively.

The final FASF block calculation is obtained as

$$\mathbf{M}_{\text{FASF}} = F^{-1}(\mathbf{M}_{\text{coarse}} \odot \mathbf{M}_{\text{fine}}). \tag{19}$$

C. MAWD Block

After stitching the raw data $\mathbf{X}$ and the mask $\mathbf{M}$ into $\mathbf{X}_{\text{cat}}$ along the channel dimension, $\mathbf{X}_{\text{cat}}$ is fed into the MAWD block to acquire deep image features. The MAWD block uses a self-attention mechanism to compute the attention weight of each channel in $\mathbf{X}_{\text{cat}}$, which enhances useful background features while suppressing the expression of anomalous features. This method can highlight the key features needed in many channels, filtering out redundant and useless information.

The specific structure of MAWD is shown in the right-hand area of Fig. 1. First, after acquiring the conditioned image $\mathbf{X}_{\text{cat}}$, the conditioned image features are compressed by a global average pooling (GAP) layer to obtain a 1-D feature vector $\mathbf{X}_{\text{gap}} \in \mathbb{R}^{H\times W\times 2C}$. The global contextual information of the conditional image can be fully extracted by this step, and its formula is

$$\mathbf{X}_{\text{gap}} = \text{GAP}(\mathbf{X}_{\text{cat}}) \tag{20}$$

where $\mathbf{X}_{\text{gap}} = 1/(H \times W) \sum_{h=1}^{H} \sum_{w=1}^{W} \mathbf{X}_{\text{cat}}(h, w, 2c)$.

After that, $\mathbf{X}_{\text{gap}}$ is input to the first fully connected (FC) layer. The FC layer enables the network to learn complex relationships between input features. The first FC layer can be seen as a transformation from a convolutional layer to high-level features. Assuming that the weight matrix of the first FC layer is $\mathbf{W}_1$ and the bias is $\mathbf{b}_1$, the output of the first FC layer is

$$\mathbf{Z}_1 = \mathbf{W}_1 \cdot \mathbf{X}_{\text{gap}} + \mathbf{b}_1. \tag{21}$$

The output $\mathbf{Z}_1 \in \mathbb{R}^{HW\times D}$ is passed through the ReLU activation function to get the result $\mathbf{A}_1 \in \mathbb{R}^{HW\times D}$. The formula is

$$\mathbf{A}_1 = \text{ReLU}(\mathbf{Z}_1). \tag{22}$$

The second FC layer can build on these high-level features to further learn complex patterns and nonlinear relationships. The output $\mathbf{A}_1$ from the ReLU activation function is fed into the second FC layer. Given the weight matrix $\mathbf{W}_2$ and the bias vector $\mathbf{b}_2$ associated with this layer, the resulting output from the second FC layer is computed as follows:

$$\mathbf{Z}_2 = \mathbf{W}_2 \cdot \mathbf{A}_1 + \mathbf{b}_2. \tag{23}$$

This hierarchical feature learning helps to improve the generalization ability and accuracy of the network, and the ReLU function is used for nonlinear activation between the two layers.

Finally, $\mathbf{Z}_2 \in \mathbb{R}^{H\times W\times 2C}$ is fed into the sigmoid layer to obtain the attentional weights of the image channels $\mathbf{W} \in \mathbb{R}^{H\times W\times 2C}$, which represent the importance of different channels in $\mathbf{X}_{\text{cat}}$. $\mathbf{W}$ is calculated by the formula

$$\mathbf{W} = \text{Sigmoid}(\mathbf{Z}_2) = \frac{1}{1 + \exp(-\mathbf{Z}_2)}. \tag{24}$$

III. Experiments and Results Analysis

In this section, the proposed FCMMA method is compared with nine state-of-the-art HAD methods on five hyperspectral datasets to evaluate its performance. In addition, the experimental setup and the qualitative and quantitative evaluation

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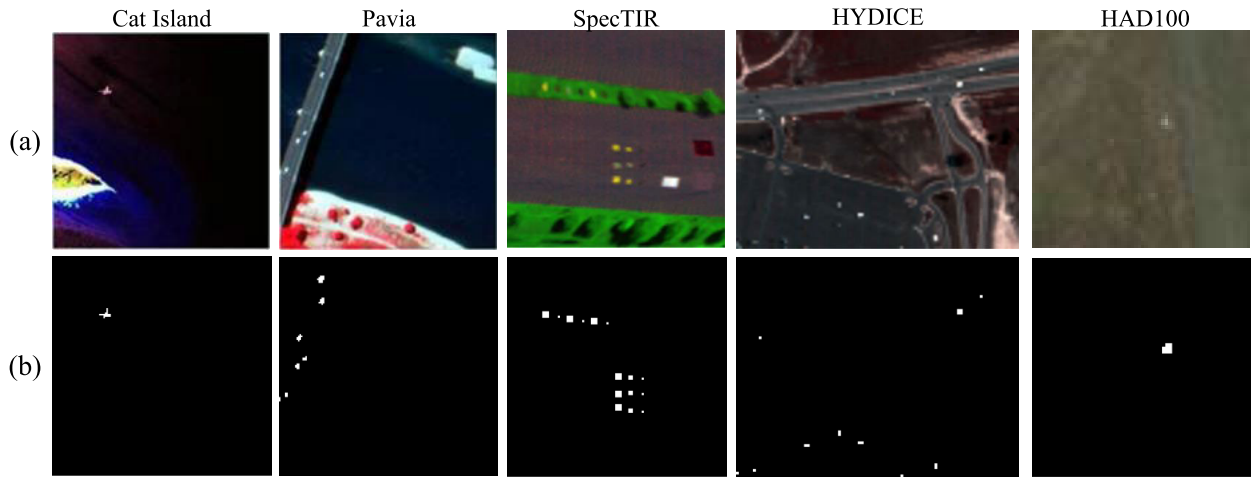


Fig. 3. Five datasets were used in the experiment. (a) Pseudo-color images of the five real HSI datasets. (b) Corresponding ground-truth images.

indicators are also introduced. Finally, the experimental results are analyzed.

A. Dataset Description

Five publicly available datasets are used to evaluate the detection performance of the proposed method. The pseudo-color and ground-truth maps for these datasets are shown in Fig. 3. The target objects were labeled using the environment for visualizing images (ENVI) software tool. These datasets contain a variety of backgrounds, such as farmland, residential areas, deserts, and beaches, with land cover types such as vegetation, soil, water, and roads. Anomalies include aircraft, vehicles, man-made fabrics, and other types.

1) Cat Island Dataset [60]: The first dataset relates to the area around Cat Island and was collected on September 12, 2010, using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. It has a spatial resolution of 17.2 m/pixel and a spatial size of 150 × 150. After excluding noise bands, 188 spectral bands were retained in the dataset for experimental analysis. The anomalies include 19 anomalous pixels on one aircraft, representing 0.08% of the total number of pixels.

2) Pavia Dataset [60]: The second dataset is concerned with the area around Pavia, mainly beach scenes, for which data were captured using the Reflectance Optical System Imaging Spectrometer (ROSIS-03) sensor. It has a spatial resolution of 1.3 m/pixel and a spatial size of 150 × 150. After excluding unsuitable bands, 102 spectral bands were retained in the dataset for experimental analysis. The major anomalies include 68 one-shot pixels, including the number of vehicles on the bridge, representing 0.30% of the total number of pixels.

3) SpecTIR Dataset [61]: The third dataset was collected in Rochester, NY, USA, with data captured using the SpecTIR Hyperspectral Airborne Rochester Experiment (SHARE). It has a spatial resolution of 1 m/pixel and a spatial size of 100 × 100. The dataset retains 360 spectral bands for experimental analysis. The anomalies include 72 anomalous pixels including man-made colored square fabrics of different sizes, representing 0.72% of the total number of pixels.

4) HYDICE Dataset [62]: The fourth dataset was collected in a suburban residential area in MI, USA, where data were captured using the Hyperspectral Digital Image Collection Experiment (HYDICE) sensor. It has a spatial resolution of 3 m/pixel and a spatial size of 80 × 100. The spectral range of this dataset is 400–2500 nm. A total of 162 bands were retained after removing absorbing regions, low signal-to-noise ratios, and poor-quality bands (1–4, 76, 87, 101–111, 136–153, and 198–210). Anomalies include 17 pixels including ten man-made vehicles. Background land cover types are vegetation, soil, water, and roads.

5) HAD100 Dataset [63]: The fifth dataset is the desert background, taken from the HAD100 dataset, with data captured by the AVIRIS-NG sensor. It has a spatial resolution of 2.3 m/pixel and a spatial size of 64 × 64. The dataset has a spectral range of 377–2500 nm with 425 raw bands. After excluding the absorbing regions and noise bands, the dataset retained 276 spectral bands for experimental analysis. The anomalies consist of eight anomalous pixels, representing 0.20% of the total number of image elements.

B. Experimental Setup

1) Comparison Methods and Parameter Settings: Nine classical or state-of-the-art HAD methods were compared with FCMMA, including RX [12], CRD [24], IEEPST [20], LREN [31], RGAE [32], Auto-AD [33], GAED [11], TAEF [34], and DFAN [37]. RX, CRD, and IEEPST are all traditional HAD methods. LREN, RGAE, Auto-AD, GAED, and DFAN are AE-based deep learning HAD methods. TAEF is a deep learning HAD method based on Transformer. Among these methods, RX and CRD are considered the classical detection methods in the field of HAD. Deep learning methods such as LREN, RGAE, Auto-AD, and GAED have been cited and followed many times in the HAD field. IEEPST, TAEF, and DFAN are the latest algorithmic advancements in the field, demonstrating their competitiveness. We set the parameters of the proposed method and other benchmarking methods based on our experience and descriptions in relevant articles.

The RX algorithm does not require tuning parameters. For the CRD, we set λ to $1e^{-6}$ and the double window size to (9, 11). For the IEEPST, the number of point sets within each

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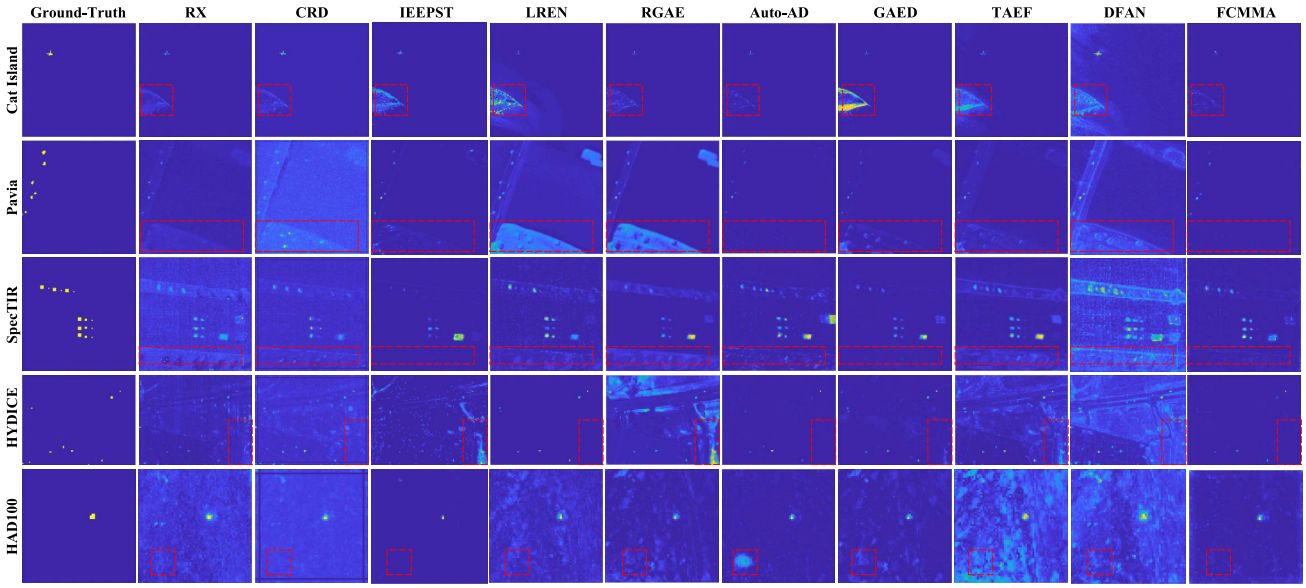


Fig. 4. Color anomaly maps based on ten methods for the five HSI datasets.

topological space is set at 140. For LREN, the value of λ is set between 10^{-1} and 0.1 with all other parameters fixed. For RGAE, set the tradeoff parameter to $1e^{-2}$ and the number of superpixels to 150. For Auto-AD, the learning rate was set to 0.01 and the number of rounds to 1001. For GAED, the number of batches is set to 10, the learning rate is set to 0.4, the window size is 7, as well and the penalty coefficient is set to 1. For TAEF, the learning rate is set to 0.005 and the number of training rounds is set to 20. For the processing of datasets I–IV, the step is set to 5 and the window size is set to 10; for the processing of dataset V, the step is set to 4 and the window size is set to 8. For DFAN, the domain radius is set to 0.3, and the minimum number of points within the domain radius is set to 3. For FCMMA, we used the adaptive moment estimation (ADAM) optimizer to optimize our model. During training, the epoch is set to 201 and the learning rate is set to $3e^{-4}$.

In terms of hardware devices, all experiments were implemented on PCs with 8.00 GB of RAM with Intel i7-8565U CPU and NVIDIA GeForce GTX 2080Ti gpu.

2) Qualitative and Quantitative Evaluation Metrics: We use qualitative metrics along with quantitative metrics to evaluate the performance of our algorithms on five datasets. The algorithms were qualitatively evaluated using color anomaly maps. The color anomaly map can visualize the effect of anomaly detection.

The algorithms were quantitatively evaluated using two-dimensional receiver operating characteristic (2-D ROC) curve [64], three-dimensional receiver operating characteristic (3-D ROC) curve, area under the ROC curve (AUC), and Box–whisker plots. The 2D-ROC and 3D-ROC show what kind of relationship the detection probability P_D presents with the false alarm probability P_F at different thresholds τ. The closer the 2D-ROC curve is to the upper left corner, the closer its AUC is to 1. In this case, the algorithm is evaluated for integrated detection performance using $\mathrm{AUC}_{(D,F)}$, and for background suppression performance using $\mathrm{AUC}_{\mathrm{BS}}$. The values of both AUC metrics follow this pattern: the larger the value, the greater the capability. Box–whisker plots can assess the separability of anomalies from the background while reflecting the original distributional characteristics of the data.

C. Comparison Results

In this section, the detection performance of the ten methods is analyzed qualitatively and quantitatively on five datasets.

1) Qualitative Evaluation: The color anomaly maps of different methods are shown in Fig. 4. For the Cat Island dataset, IEEPST, LREN, GAED, TAEF, and DFAN did not effectively suppress the background, and false alarms can still be seen in the lower left corner of their detection images. For other methods, RX, CRD, and RGAE have better background suppression effects than the first five methods, but there are also significant false alarms. The Auto-AD method excels in background suppression but exhibits less distinct detection of anomalous targets. It can be observed that FCMMA effectively suppresses background pixels while accurately detecting abnormal pixels.

For the Pavia dataset, there are a large number of false alarms below the color anomaly maps for RX, CRD, LREN, RGAE, and DFAN. There are fewer false alarms for IEEPST, GAED, and TAEF, but their presence can still be seen at the bottom of the images. The background suppression ability of Auto-AD and FCMMA is very excellent, with almost no false detected pixels on the color anomaly map, but FCMMA detects more abnormal pixels compared to Auto-AD.

For the specTIR dataset, it can be seen that RX, CRD, LREN, RGAE, Auto-AD, TAEF, and DFAN have poor background suppression capabilities. Especially in the horizontal stripes above and below the color anomaly maps, none of the above algorithms were able to effectively suppress them. The background of IEEPST, GAED, and FCMMA is relatively clean, indicating an excellent background suppression effect on this dataset. At the same time, anomalous pixels are always

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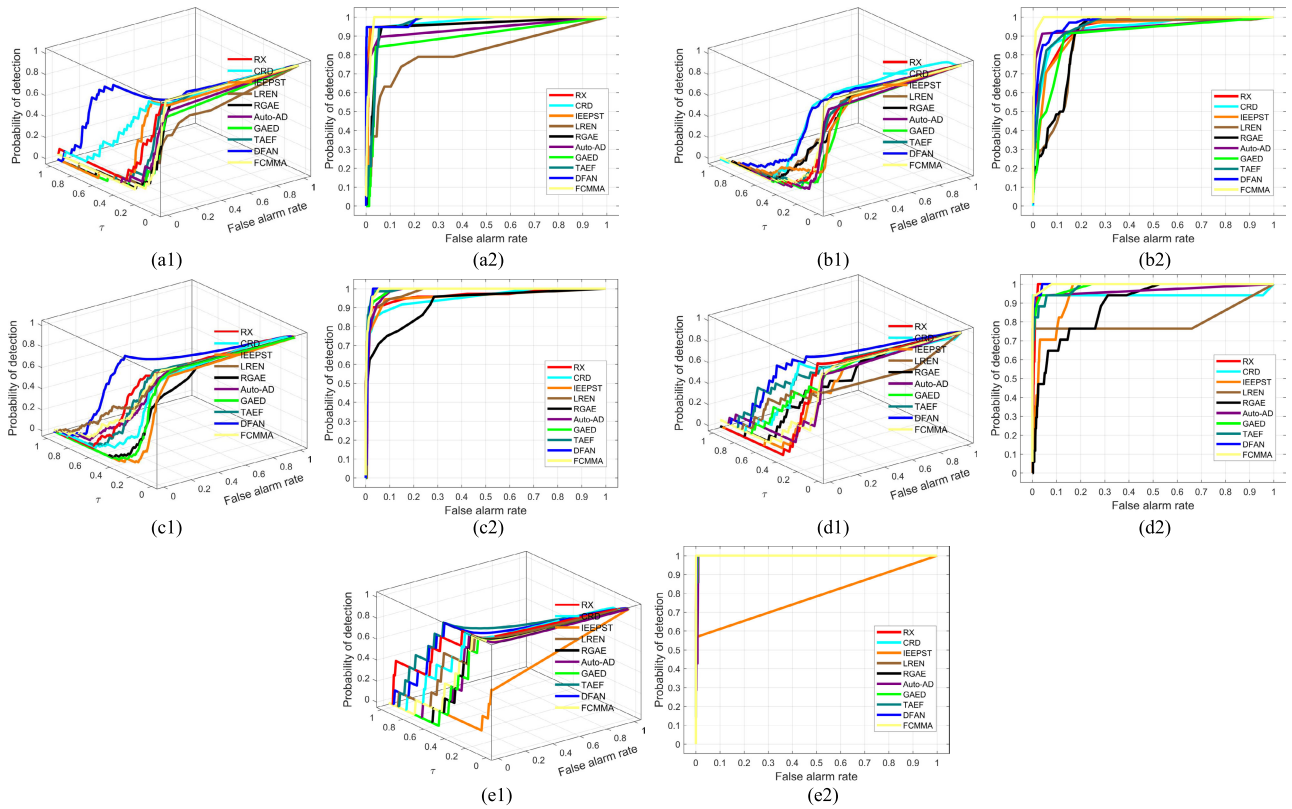


Fig. 5. ROC curves based on ten methods for the five HSI datasets. (a) Cat Island. (b) Pavia. (c) SpecTIR. (d) HYDICE. (e) HAD100. (a1)–(e1) 3D-ROC curves. (a2)–(e2) 2D-ROC curves.

detected more highlighted and complete on the detection map of FCMMA.

For the HYDICE dataset, RX, CRD, IEEPST, RGAE, TAEF, and DFAN did not effectively suppress the background, and there are still noticeable background traces on the detection images. On the contrary, LREN, Auto-AD, GAED, and FCMMA all have excellent background suppression effects on this dataset and can comprehensively detect abnormal pixels.

For the HAD100 dataset, there are obvious background residues in the color anomaly maps of RX, CRD, LREN, RGAE, GAED, TAEF, and DFAN, and none of the above algorithms have achieved good background suppression on this dataset. Auto-AD suppresses a large number of background pixels, but there is a large residual false alarm in the lower left corner. On the contrary, IEEPST and FCMMA accurately detected abnormal pixels while effectively suppressing background pixels.

2) Quantitative Evaluation: The 3D-ROC curves and 2D-ROC curves of ten methods on five datasets are shown in Fig. 5(a1)–(e1) and (a2)–(e2). Among them, the 3D-ROC curves of different datasets in Fig. 5(a1)–(f1) intuitively show the detection performance of different algorithms at different thresholds τ. It can be seen that the 3D-ROC curve of FCMMA performs well in multiple dimensions, and the shape and coverage of the curve demonstrate the efficiency and accuracy of the model in anomaly detection from multiple angles. The 2D-ROC curves for detection methods such as IEEPST, LREN, GAED, and DFAN do not always reach the top-left corner. This suggests there might be problems with their accuracy in detection, parameter adjustment, or feature selection, resulting in an imperfect distinction between true positives and false positives. In Fig. 5(a2)–(e2), 2D-ROC curves of FCMMA are always closer to the top left corner. This means that FCMMA has a better overall detection level compared to other methods.

The $\mathrm{AUC}_{(D,F)}$ values of all methods on five datasets are shown in Table I. It can be seen from the quantitative data that RX, Auto-AD, and TAEF have high average values, which are all 0.9700 or above. Among them, it can be seen that the $\mathrm{AUC}_{(D,F)}$ value of Auto-AD is relatively stable, which indicates its good robustness. The RX, LREN, and RGAE algorithms demonstrate instability, with their detection values exhibiting considerable variation across different datasets. This is likely due to their poor adaptability to varying data distributions. However, FCMMA has the highest stable $\mathrm{AUC}_{(D,F)}$ value on all datasets, which shows that it has the best detection performance.

The $\mathrm{AUC}_{\mathrm{BS}}$ values for all methods on the five datasets are shown in Table II. Numerically, CRD, LREN, and RGAE exhibit lower $\mathrm{AUC}_{\mathrm{BS}}$ values across five datasets, indicating weaker average background suppression. In stark contrast, FCMMA excels in background suppression, achieving the highest $\mathrm{AUC}_{\mathrm{BS}}$ values on all datasets, each exceeding 0.9700. This robust performance of FCMMA is likely due to its more effective handling of background complexity and noise. In terms of mean value, FCMMA is only slightly higher than the Auto-AD algorithm, but it has a great background suppression advantage over the other algorithms. Overall, the FCMMA has very good background suppression, which makes it better at highlighting anomalous targets.

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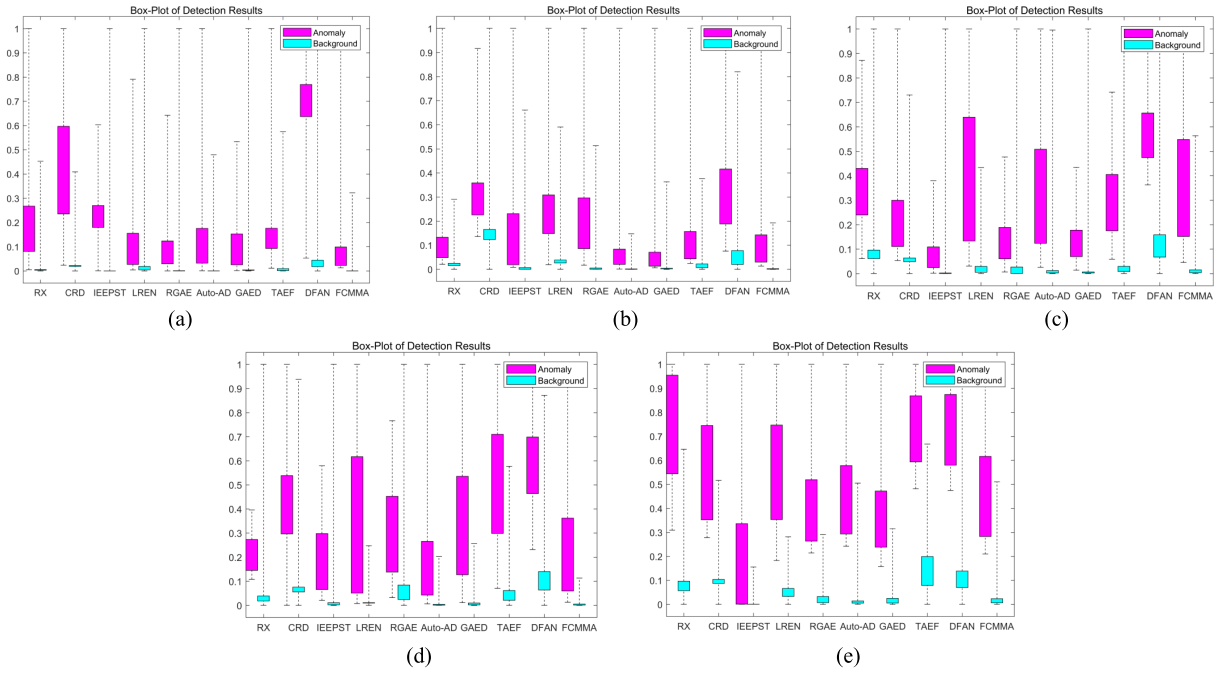


Fig. 6. Box–whisker plots based on ten methods for the five HSI datasets. (a) Cat Island. (b) Pavia. (c) SpecTIR. (d) HYDICE. (e) HAD100.

TABLE I
$AUC_{(D,F)}$ VALUES OF ALL METHODS ON FIVE DATASETS

Datasets	RX	CRD	IEEPST	LREN	RGAE	Auto-AD	GAED	TAEF	DFAN	FCMMA
Cat Island	0.9807	0.9893	0.9646	0.8310	0.9392	0.9812	0.9223	0.9622	0.9893	**0.9905**
Pavia	0.9538	0.9440	0.9331	0.8905	0.9042	0.9834	0.9372	0.9637	0.9740	**0.9960**
SpecTIR	0.9655	0.9563	0.9568	0.9830	0.9347	0.9831	0.9918	0.9928	0.9914	**0.9933**
HYDICE	0.9932	0.9406	0.9511	0.8002	0.8907	0.9904	0.9844	0.9829	0.9967	**0.9972**
HAD100	0.9994	0.9995	0.8104	0.9993	0.9993	0.9958	0.9990	0.9994	0.9990	**0.9996**
Average	0.9785	0.9659	0.9232	0.9008	0.9336	0.9868	0.9669	0.9802	0.9901	**0.9953**

Fig. 6(a)–(e) shows the box–whisker plots of methods on five datasets. In the figure, red represents anomalous pixels and blue represents background pixels. The distance between the red box and the blue box represents the degree of separation between the anomaly and the background. As shown in Fig. 6(a), the two boxes of LREN are close to each other, which shows that it cannot well separate anomalous pixels from background pixels. For each dataset, the separation of the two boxes of the FCMMA method is clear, which can be seen that it has a good effect of separating background and anomaly. The shorter the box, the fewer the pixels. Fig. 6(e) shows that TAEF and DFAN have extended background boxes, indicating poor background suppression on the HAD100 dataset. On the contrary, each blue background box of FCMMA is very short, which indicates that FCMMA has a good background suppression effect on five datasets. In particular, as shown in Fig. 6(d), FCMMA even suppresses almost all background pixels.

In summary, on color anomaly maps, it can be seen that FCMMA has a more intuitive detection with a cleaner background. On the ROC curves, it can be seen that the FCMMA curve is always closest to the upper left corner and outperforms the other curves in different dimensions. Moreover, at two AUC values, FCMMA visually shows the combined detection ability and background suppression capability that outperforms other algorithms. On box–whisker, it is always evident with both boxes that FCMMA has better separation of anomalous pixels from background pixels and better background suppression. Based on the results of the above qualitative and quantitative analyses, FCMMA has a superior anomaly detection capability compared to other algorithms.

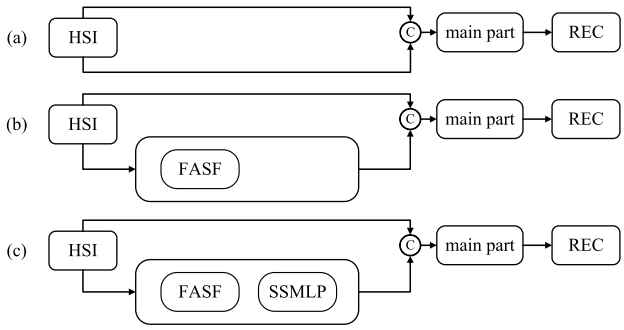


Fig. 7. Detailed schematic of the ablation steps. (a) Base. (b) Base + FASF. (c) Base + FASF + SSMLP.

D. Ablation Study

This section conducts ablation experiments on five datasets to verify the effectiveness of three parts of the proposed method, which includes base, base + FASF, and base + FASF + SSMLP. A schematic of the models for the ablation

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TABLE II
AUC_{BS} VALUES OF ALL METHODS ON FIVE DATASETS

Datasets	RX	CRD	IEEPST	LREN	RGAE	Auto-AD	GAED	TAEF	DFAN	FCMMA
Cat Island	0.9732	0.9673	0.9534	0.8075	0.9210	0.9751	0.9032	0.9440	0.9452	**0.9845**
Pavia	0.9306	0.7972	0.9245	0.8248	0.8633	0.9779	0.9272	0.9471	0.9191	**0.9902**
SpecTIR	0.8824	0.8988	0.9476	0.9608	0.9148	0.9647	0.9835	0.9670	0.8637	**0.9785**
HYDICE	0.9601	0.8727	0.9275	0.7886	0.8128	0.9845	0.9751	0.9336	0.8885	**0.9912**
HAD100	0.9199	0.9060	0.8052	0.9460	0.9733	0.9799	0.9788	0.8499	0.8857	**0.9803**
Average	0.9332	0.8884	0.9116	0.8655	0.8970	0.9764	0.9536	0.9283	0.9004	**0.9849**

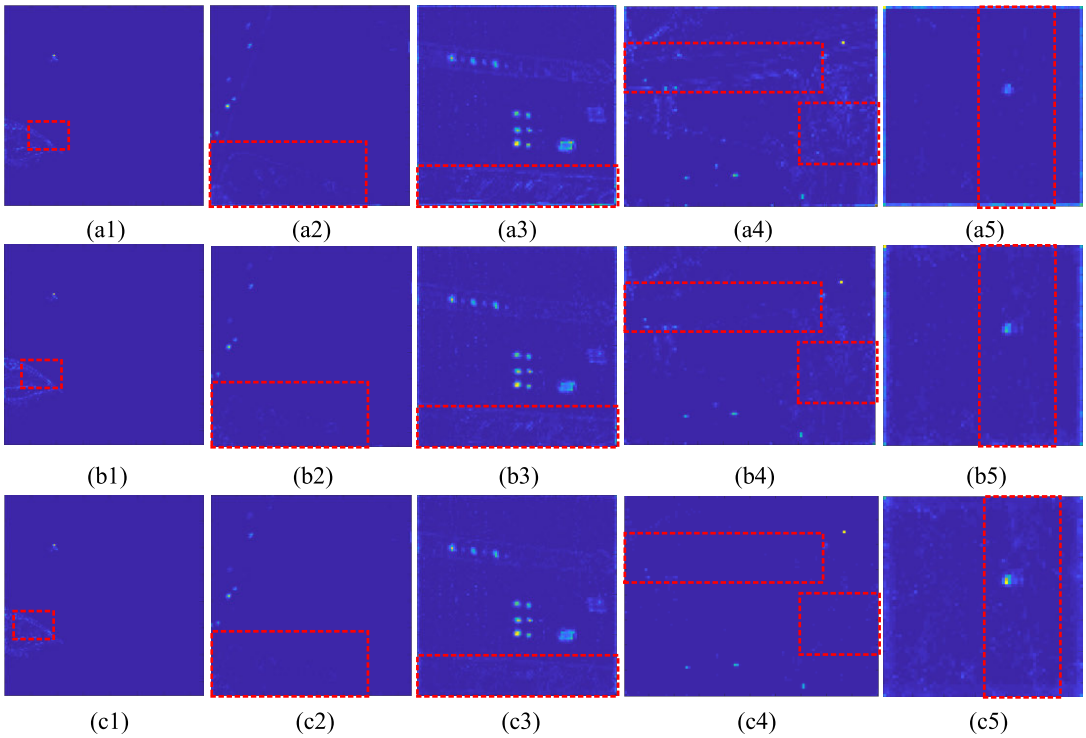


Fig. 8. Color anomaly maps of five HSI datasets are based on different modules. (a)–(c) Color anomaly maps for base, base + FASF, and base + FASF + SSMLP, respectively. (a1)–(c1) Cat Island. (a2)–(c2) Pavia. (a3)–(c3) SpecTIR. (a4)–(c4) HYDICE. (a5)–(c5) HAD100.

experiments carried out by different modules is given in Fig. 7. First, as shown in Fig. 7(a), the validity of the BASE structure of our proposed framework will be verified. To ensure that the three steps have the same frame structure, the lower branch is directly connected to the splicer. Second, as shown in Fig. 7(b), the proposed FASF module is added to the BASE, and the effectiveness of FASF in terms of anomaly detection effect is verified. Where to ensure the integrity of the CMASK structure, the right-hand side of this step uses a normal MLP as an alternative. Finally, the SSMLP block was added to the FASF as shown in Fig. 7(c) to verify its validity. All the above experiments were done on the same device using the same parameter settings. The results of the ablation experiments were assessed using the qualitative color anomaly maps shown in Fig. 8 and the quantitative $AUC_{(D,F)}$ values shown in Table III.

1) Effectiveness of the Proposed Framework: As shown in the colored anomaly diagram in the top row of Fig. 8, the algorithm still detects the anomalous pixels when only the proposed framework is used, and there are no exceptionally large and noticeable residual backgrounds in the whole image, but only a few local residual backgrounds. For example, there are still localized background residues in Fig. 8(a2) and (a4). However, overall the proposed framework still has initially a better anomaly detection effect. As shown by the quantified AUC values in Table III, the AUC values of the BASE model can reach more than 0.9400 on all six datasets and can even reach more than 0.9800 on Cat Island, Pavia, and SpecTIR. Based on the qualitative and quantitative analyses, it can be concluded that the proposed framework has more excellent detection results.

2) Effectiveness of the Proposed FASF: As shown in the colored anomaly diagram in the second row of Fig. 8, the background suppression ability of the model is improved with the addition of the FASF module. Based on the image comparison of the part of the figure circled by the red box, it can be seen that the partially unsuppressed background in BASE becomes less visible. For example, the resultant images in Fig. 8(b2) and (b4) with the addition of FASF are much sharper and cleaner, with more adequate background suppression in the portion of the red circle, than their images when using base-only. As shown by the quantified AUC values in Table III, the AUC values at base + FASF are higher than the AUC values when using base only, except for the HYDICE dataset. The reason for the slightly lower AUC value in the HYDICE dataset was analyzed to be that some of the anomalous pixels were suppressed when the background was

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TABLE III
ABLATION RESULTS FOR $AUC_{(D,F)}$ VALUES ON FIVE DATASETS

Datasets	base	base+FASF	base+FSAF+SSMLP
Cat Island	0.9824	0.9881	**0.9905**
Pavia	0.9883	0.9933	**0.9960**
SpecTIR	0.9831	0.9924	**0.9933**
HYDICE	0.9950	0.9912	**0.9972**
HAD100	0.9614	0.9910	**0.9996**

suppressed using FASF, leading to this result. Overall, however, FASF is still a very effective part of the CMASK, which effectively suppresses anomalous pixels in the mask, resulting in fewer residual background pixels in the difference map.

3) Validity of the Proposed SSMLP: As shown in the colored anomaly diagram in the last row of Fig. 8, the background suppression ability of the model and the anomaly highlighting ability are simultaneously improved after the addition of the SSMLP module. Sections where changes are evident are circled in red. For example, in Fig. 8(c2)–(c4), there is still a small portion of the background remaining in the resultant plots when using the base + FASF model, whereas with the addition of the SSMLP block, the background is almost completely suppressed, and the portion of the original false positives is almost difficult to see. Meanwhile, in Fig. 8(c4) and (c5), the anomalous pixels, which were still relatively dark, become more prominently highlighted. Fig. 8(a5)–(c5) show that the FASF block reduces exaggerated edges, and the SSMLP block nearly eliminates edge misdetection, which is almost imperceptible to the naked eye. This enhancement is due to SSMLP's ability to increase the expressiveness of background pixels and to correct misdetected edges during the reconstruction process of the BASE and FASF blocks. Meanwhile, according to the quantitative results in Table III, it can be seen that the AUC values of FCMMA on all datasets are improved after the addition of SSMLP, and all of them reach above 0.9900. The effectiveness of SSMLP in background suppression and anomalous target highlighting was demonstrated based on the above analysis of qualitative and quantitative results.

In summary, the effectiveness of all three modules of FCMMA was verified through ablation experiments. First, it is verified that the proposed framework foundation initially has better anomaly detection. Second, the inclusion of the FASF module allows for fuller background suppression of the assay. Finally, the inclusion of the SSMLP module further suppresses the background while highlighting anomalous targets. Meanwhile, as shown in Table III, the AUC values under all three components are high and gradually increase with the increase of modules. From the results of the above qualitative and quantitative analyses, it can be seen that all three modules can effectively improve the performance of the proposed method.

IV. CONCLUSION

This article proposed the FCMMA method for HAD. It addresses the issue of rough suppression of anomalous pixels during AE reconstruction, ensuring that background pixels are accurately preserved. Specifically, the FASF module is designed to achieve coarse and fine suppression of anomalies in the frequency domain. FASF utilizes Fourier techniques to effectively manage low-frequency background and high-frequency anomaly components. Additionally, the CMASK network is devised to generate CMASKs for enhanced anomaly suppression. The CMASK network employs the FASF module and the SSMLP module to further enhance the transformation and representation capabilities of the generated masks. Extensive experiments indicate that the FCMMA method outperforms nine HAD methods on five datasets, validating its effectiveness and superiority.

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